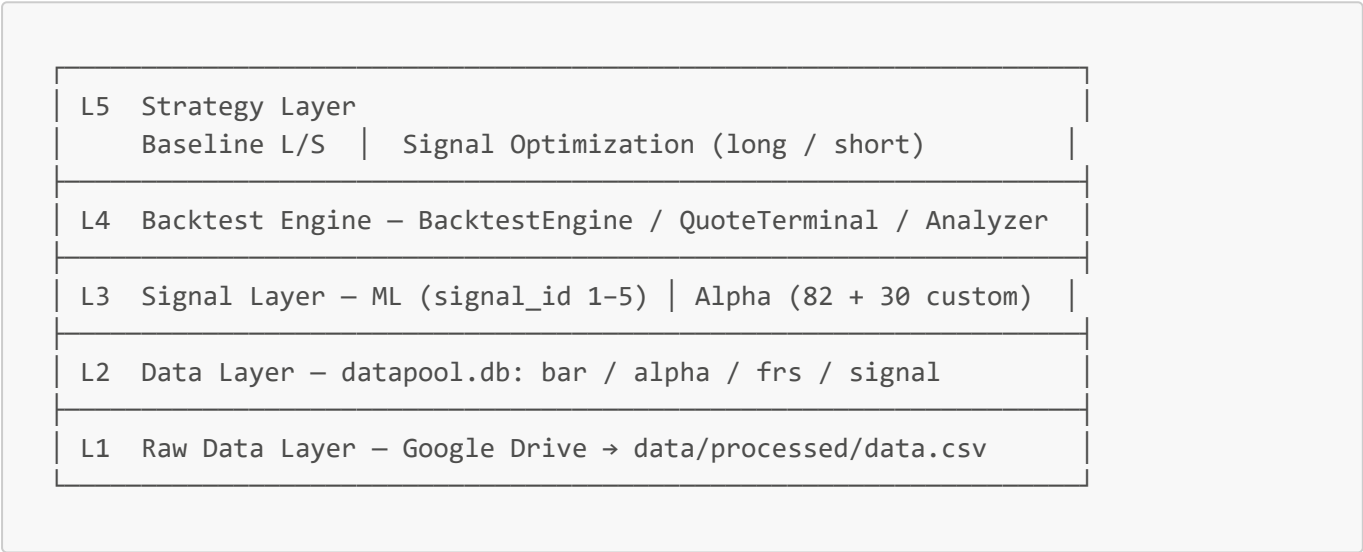


QuantLab Research Report — 2026-05-13

Branch: `feature-infra`
In-Sample (IS): 2021-03-03 → 2024-12-31 · 200 weekly periods
Out-of-Sample (OOS): 2025-01-01 → 2026-03-01 · 61 weekly periods
Universe: 11 SPDR Sector ETFs · Zero fees, zero slippage unless stated

1. System Architecture

QuantLab is a five-layer quantitative research stack for weekly sector ETF rotation. Each layer has a single well-defined responsibility; data flows strictly upward from raw files to strategy execution.



Module Responsibilities

Module	Answers	Key Output	Constraint
Signal	What to trade	<code>OrderedDict[ticker → score]</code>	Read-only; no account access
Strategy	How to trade	List of <code>Action</code> objects	Proposes only; no risk enforcement
Risk	Whether to trade	Filtered / clamped action list	Final gatekeeper; all constraints here

Key Design Decisions

Decision	Rationale
Wednesday-close signal / Thursday-close execution	Eliminates look-ahead bias; compresses weekend gap to a single overnight hold
<code>QuoteTerminal</code> as the single time truth	All modules share the same data cut-off via <code>bind(terminal)</code> , preventing date drift
<code>alpha_110</code> hard-wired short filter	Prevents shorting ETFs in upward 12-week momentum trends; non-negotiable risk rule

Decision	Rationale
Softmax allocation (no hard top-k)	Preserves gradient flow for Step 3 neural-network differentiable backtest training
Soft NAV floor $\epsilon = 1e-6$	Numerical continuity for differentiable backtest paths; prevents $\log(0)$
Equal-weight as mandatory Sharpe floor	Forces signals to prove cross-sectional information content, not just market beta

Project Directory

```

LAB/
├── src/QuantLab/
│   ├── alpha/                # Alpha registration & compute
│   ├── frs/                  # FRS registration & compute
│   ├── signal/               # Signal registration
│   │   └── ml/                # ML signal implementations (signal_1-5)
│   ├── backtest/
│   │   ├── engine.py         # Main simulation loop
│   │   ├── quote_terminal.py # Unified data access (anti-look-ahead)
│   │   ├── analyzer.py       # Metrics & visualization
│   │   ├── signal/           # AlphaBacktestSignal, MLBacktestSignal
│   │   └── strategy/         # SiganlOptimizationStrategy,
├── BaselineStrategy
│   │   └── risk/              # BaselineRisk, DebugRisk
│   └── utils/                 # db / data_loader / load_ml_input / config
├── backtests/
│   ├── baseline/
│   │   ├── no_trans_cost/    # zero-fee / zero-slippage runs
│   │   │   ├── in_sample/alpha/
│   │   │   ├── out_sample/alpha/
│   │   │   ├── in_sample/ml_signal/
│   │   │   └── out_sample/ml_signal/
│   │   └── trans_cost/       # friction-cost runs (planned)
│   └── signal_optimization/
│       ├── in_sample/long_only/ | in_sample/short_only/
│       └── out_sample/long_only/ | out_sample/short_only/
├── data/processed/           # data.csv (main input)
├── debrief/                   # Periodic team reports
├── docs/                      # Design documents (AGENT.md + 00-04)
├── research/                  # Per-contributor notes & archives
│   ├── 20260426_Team_Early-Baseline/ # ARCHIVED: original prototype
│   ├── 20260426_David_Alpha-signals/
│   ├── 20260426_Simon_ML Infra/
│   ├── 20260509_Simon_Signal_Infra/
│   └── 20260513_Simon_Strategy/
└── scripts/                   # dataset_builder.ipynb

```

2. Data Layer

Universe

11 SPDR Sector ETFs: XLB, XLC, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLRE, XLY

Benchmark series (read-only): SPY, SPX, VIX, US10Y

Database Schema — datapool.db

Table	Content	Frequency	Key Columns
daily_bar	OHLCV + TRI, all 15 assets	Daily	ticker, date, open, high, low, close, volume, tri
weekly_bar	Wednesday-close resampled, all 15 assets	Weekly	ticker, date, open, high, low, close, volume, tri
weekly_alpha	Alpha factor values, ETFs only	Weekly	ticker, date, alpha_id, value, group (wq101 / andy)
weekly_frs	FRS1/2/3 forward return scores, ETFs only	Weekly	ticker, date, frs_id, value
weekly_signal	ML / alpha / composite signals, ETFs only	Weekly	ticker, date, signal_id, value

FRS Label Definitions

Code	Name	Formula	Use
FRS1	4-week total return	$(P_4 - P_0) / P_0$	Signal 2 label; used in rank-average ensemble
FRS2	Sharpe proxy	$\text{avg}(r_1 \dots r_4) / \text{std}(r_1 \dots r_4)$	Signal 5 label (MLP)
FRS3	Vol-penalized return	$\text{FRS1} - 2.0 \times \text{std}(r_1 \dots r_4)$	Signals 1, 3, 4 label (LightGBM / XGBoost / PCA-Ridge)

Data Pipeline

```
Google Drive → data/processed/data.csv
↓
dataset_builder.ipynb → datapool.db (daily_bar, weekly_bar)
↓ alpha_metrics.py
weekly_alpha (WQ101 + Andy factors, IDs 1-101 and 108-137)
↓ frs_metrics.py
weekly_frs (FRS1, FRS2, FRS3)
↓ signal_metrics.py + ML training
weekly_signal (ML signal_id 1-5)
```



QuoteTerminal → BacktestEngine → Strategy / Risk

Developer extension rule: add a decorated function to `*_metrics.py` and re-run the `save_*` function. Registry rows are append-only; fact tables are refreshed per run.

3. Signal Pool

3.1 WQ101 Alpha Factors

Implemented from Kakushadze (2015) *101 Formulaic Alphas*. All factors stored in `weekly_alpha` with `group='wq101'`.

Group	Count	Note
Group A — Fully implementable	52	All required fields (OHLCV + volume) available
Group B — vwap approximated	30	$\text{vwap} \approx (\text{O} + \text{H} + \text{L} + \text{C}) / 4$
Not implementable	19	Requires <code>IndNeutralize</code> or market-cap data — unavailable for ETFs

Top alphas by mean IC: Alpha#50 (0.042), #3 (0.034), #41 (0.034), #24 (0.033), #98 (0.031).

3.2 Andy Alpha Factors (Custom Group)

30 custom alpha factors (IDs 108–137), stored with `group='andy'`. Designed to capture trend, risk-adjusted momentum, liquidity, and relative-strength effects specific to the sector ETF universe.

Family	IDs	Description
Price Momentum	108–112	Cumulative returns over 1w, 4w, 8w, 12w, 26w windows
Risk-Adjusted Momentum	113–117	Return / rolling σ over matched windows
Realized Volatility	118–122	Rolling σ of weekly returns (4w–52w)
Log Dollar Volume	123–124	$\log(\text{price} \times \text{volume})$ rolling 4w / 12w
Amihud Illiquidity	125–126	$ \text{return} / \text{dollar_volume}$ rolling 4w / 12w
Drawdown from Peak	127–128	Current price / rolling max over 12w / 52w
Distance from Moving Average	129–131	$(\text{Price} - \text{SMA}) / \text{SMA}$ over 4w, 12w, 26w
SPY Relative Strength	132–134	ETF return – SPY return over 4w, 12w, 26w
Negated SPY Correlation	135–137	$-\text{corr}(\text{ETF}, \text{SPY})$ over 12w, 26w, 52w

Note: `alpha_110` (12-week cumulative return) is hard-wired as the short-entry filter in `BaselineRisk` and is not used as a ranking signal.

3.3 ML Signals

Five ML signals trained on the ETF universe and stored in `weekly_signal` (signal_id 1–5). All models use walk-forward cross-validation on the IS period.

signal_id	Model	Label	OOS Baseline Sharpe
1	LightGBM	FRS3	−1.571
2 ★	Ensemble (Rank Average)	FRS1	0.862
3	XGBoost	FRS3	−0.731
4	PCA + Ridge	FRS3	0.267
5	MLP	FRS2	−0.087

4. Backtest Engine

4.1 Four-Module Design

Module	Role	Key API
<code>BacktestEngine</code>	Simulation loop — advances clock, coordinates modules, applies fills	<code>engine.run()</code> , <code>engine.evaluate()</code>
<code>QuoteTerminal</code>	Single source of truth for time + data (anti-look-ahead)	<code>terminal.at(day)</code> , <code>terminal.signals()</code>
<code>Signal / Strategy / Risk</code>	Signal → ranking → position intent → risk filter	<code>analyze()</code> , <code>on_ranking()</code> , <code>on_action()</code>
<code>BacktestAnalyzer</code>	Performance metrics + multi-panel dashboard (NAV vs SPY)	<code>.json</code> , <code>.md</code> , <code>.csv</code> , <code>.png</code> artifacts

4.2 Simulation Flow (per weekly period)

Step	Who	What happens
1	BacktestEngine	<code>terminal.at(today)</code> — advances data cut-off to current Wednesday close
2	Signal	<code>analyze()</code> — reads history via QuoteTerminal, returns <code>OrderedDict[ticker → score]</code>
3	Strategy	<code>on_ranking(ranking)</code> — converts scores to a list of proposed Actions with sizes
4	Risk	<code>on_action(actions)</code> — filters, clamps, or blocks actions; updates internal risk state
5	Trader	<code>on_action(filtered)</code> — converts Actions to Trades at Thursday-close prices
6	Account	<code>on_trades(trades)</code> — updates cash, positions, NAV; records snapshot

Step	Who	What happens
7	BacktestAnalyzer	<code>evaluate()</code> — writes <code>*_metrics.json</code> , <code>*_metrics.md</code> , <code>*_value_history.csv</code> , <code>*_all_in_one_panel.png</code>

4.3 Anti-Look-Ahead Architecture

Core contract: inside `Signal.analyze()`, `Strategy.on_ranking()`, and `Risk.on_action()`, all data access goes through `self.terminal.*()` queries. Every method on `QuoteTerminal` applies a hard filter `date ≤ terminal.day` before returning any row. There is no path for future data to leak in.

4.4 QuoteTerminal API

Method	Returns	Data cut-off
<code>terminal.at(day)</code>	Sets simulation date	N/A (setter)
<code>terminal.day</code>	Current simulation date	N/A (getter)
<code>terminal.quote(ticker)</code>	Single <code>weekly_bar</code> row for today	Exactly <code>terminal.day</code>
<code>terminal.today_etfs()</code>	ETF cross-section for today	Exactly <code>terminal.day</code>
<code>terminal.etfs()</code>	ETF <code>weekly_bar</code> history (wide)	<code>≤ terminal.day</code> inclusive
<code>terminal.signals()</code>	<code>weekly_signal</code> wide history	<code>≤ terminal.day</code> inclusive
<code>terminal.benchmarks()</code>	SPY / SPX / VIX / US10Y history	<code>≤ terminal.day</code> inclusive

5. Signal Optimization Framework

The signal optimization pipeline evaluates individual signals by mapping cross-sectional scores to portfolio weights via a softmax allocator, then backtesting frictionlessly.

5.1 Softmax Allocator

Long (LP): $w_i = \exp(s_i) / \sum_j \exp(s_j)$

Short (SP): $w_i = \exp(-s_i) / \sum_j \exp(-s_j)$ [negated scores → low-score ETFs get short weight]

- **Long Power (LP):** softmax on raw scores — higher score → more weight → bets on outperformers
- **Short Power (SP):** softmax on negated scores — lower score → more short weight → bets against underperformers

5.2 Equal-Weight Baseline (Step 0)

Identical scores imply $w_i = 1/N$. This is the mandatory Sharpe floor in the signal optimization pipeline.

Period	LP Sharpe	LP Ann. Return	SP Sharpe	SP Ann. Return
IS	0.690	9.73%	-0.690	-10.94%
OOS	1.537	16.40%	-1.537	-15.01%

Short-only structural loss: all SP Sharpes are ≤ 0 . A pure short-only book on US sector ETFs is structurally loss-making in a secular bull market.

5.3 Optimization Steps

Step	What is optimized	Method	Must beat	Status
Step 0	Equal-weight baseline	—	—	✓ Complete
Step 1	Single signal / alpha score stream	IC pre-screen + grid search	Step 0 Sharpe	✓ Complete
Step 2	Linear multi-signal mix $\sum \alpha_k g^{(k)}$	Bayesian opt + walk-forward CV	Steps 0–1	🔄 Next
Step 3	Neural network parameters $\theta \rightarrow$ per-ETF logits	SGD / Adam + differentiable backtest	Steps 0–2	⌚ Planned

6. Baseline Long-Short Strategy

6.1 Position Structure

Parameter	Value	Note
Long positions	Top 3 by signal rank	+33.3% weight each
Short positions	Bottom 3 by signal rank	-33.3% weight each
Target gross exposure	200%	NORMAL state
Target net exposure	~0%	Market-neutral
Rebalance frequency	Weekly	Wednesday signal / Thursday execute

6.2 Short-Entry Filter

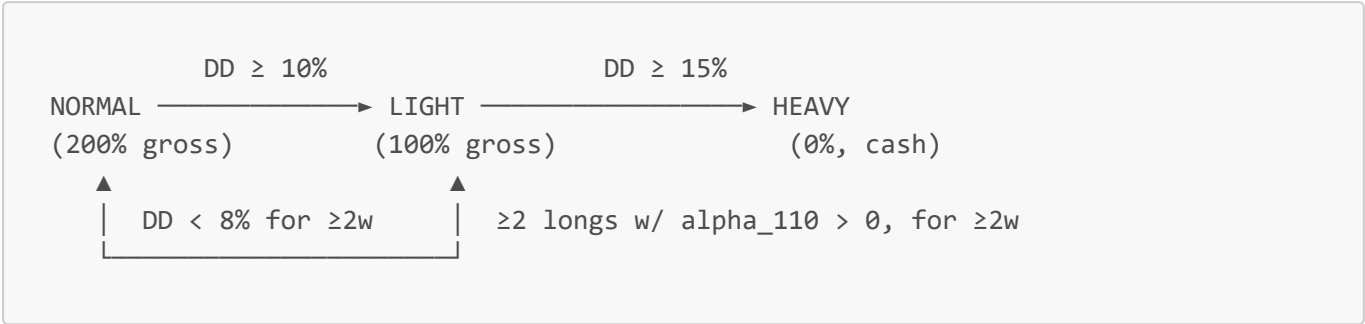
A short position is only opened if $\text{alpha}_{110} < 0$ (12-week cumulative return is negative). This prevents shorting ETFs with positive intermediate-term momentum. If fewer than 3 ETFs satisfy the filter, the short book shrinks proportionally.

6.3 Rank Stickiness

Existing holdings are retained without rebalancing if the ETF's current rank stays within a tolerance band of the cutoff, avoiding whipsaw from minor rank fluctuations.

Side	Retained if
Long	Rank $\leq n_{\text{long}} + 2$ (i.e., ≤ 5 out of 11)
Short	Rank $\geq n_{\text{total}} - n_{\text{short}} + 1 - 2$ (i.e., ≥ 7 out of 11)

6.4 3-Tier Risk State Machine (BaselineRisk)



Transition	Trigger	Recovery condition
NORMAL → LIGHT	Portfolio drawdown $\geq 10\%$	DD $< 8\%$ for ≥ 2 consecutive weeks
LIGHT → HEAVY	Portfolio drawdown $\geq 15\%$	—
LIGHT → NORMAL	—	DD $< 8\%$ for ≥ 2 consecutive weeks
HEAVY → LIGHT	—	≥ 2 proposed longs with $\alpha_{110} > 0$, for ≥ 2 consecutive weeks

7. Backtest Results

Parameters: IS = 2021-03-03 → 2024-12-31 (200 weekly periods) | OOS = 2025-01-01 → 2026-03-01 (61 weekly periods) | Initial NAV = 10,000 | Zero fees and slippage

7.1 Best-Pick Summary

Suite	Split	Best Pick	Sharpe	Ann. Return	Max DD
BL — Alpha (40 IDs)	IS	Alpha#66	0.967	8.90%	−12.83%
BL — Alpha (40 IDs)	OOS	Alpha#23 ★	1.819	19.34%	−6.47%
BL — ML (signals 1–5)	IS	Signal 1 LightGBM_frs3	0.601	5.09%	−13.71%
BL — ML (signals 1–5)	OOS	Signal 2 Ensemble_RankAvg ★	0.862	10.24%	−4.59%
LP — Alpha (signal-opt)	IS	Alpha#24	1.122	21.38%	—
LP — Alpha (signal-opt)	OOS	Alpha#57 ★	2.174	30.06%	—
LP — ML Step 1	IS	Signal 5 MLP_frs2	0.705	9.98%	—

Suite	Split	Best Pick	Sharpe	Ann. Return	Max DD
LP — ML Step 1	OOS	Signal 2 Ensemble_RankAvg ★	1.730	19.12%	—

7.2 Signal Optimization — Alpha Grid (LP / SP / BL)

Alpha	LP-IS Sharpe	LP-OOS Sharpe	SP-IS Sharpe	SP-OOS Sharpe	BL-IS Sharpe	BL-OOS Sharpe
EW Step 0	0.690	1.537	−0.690	−1.537	—	—
#19	0.564	2.014	−0.593	−1.104	−0.293	−0.618
#23	0.666	1.829	−0.787	−0.522	0.432	1.819
#24	1.122	2.055	−0.420	−2.339	0.119	−0.375
#57	0.691	2.174	−0.559	−1.136	0.352	1.315
#66	0.744	1.532	−0.630	−1.475	0.967	−0.483

7.3 Baseline L/S — Alpha Screening (selected)

In-Sample (2021-03-03 → 2024-12-31):

alpha_id	Sharpe	Ann. Return	Ann. Vol	Max DD	Turnover Ann.
66 ★	0.967	8.90%	9.26%	−12.83%	4535%
101	0.923	10.37%	11.39%	−17.27%	3134%
32	0.684	7.43%	11.44%	−13.05%	2833%
128 (Andy)	0.609	6.60%	11.60%	−13.89%	3151%
57	0.352	2.44%	7.67%	−15.68%	2124%
23	0.432	3.08%	7.73%	−17.81%	1834%

Out-of-Sample (2025-01-01 → 2026-03-01):

alpha_id	Sharpe	Ann. Return	Ann. Vol	Max DD	Turnover Ann.
23 ★	1.819	19.34%	10.00%	−6.47%	4787%
51	1.478	15.09%	9.84%	−5.84%	4305%
37	1.463	16.38%	10.77%	−5.78%	3923%
57	1.315	14.40%	10.67%	−7.40%	5425%
10	1.090	11.58%	10.57%	−6.90%	4976%
66	−0.483	−5.25%	10.11%	−9.27%	4542%

IS best (Alpha#66) collapses OOS (Sharpe -0.483). Weak IS/OOS rank correlation across all 40 alphas confirms high overfitting risk in IS-only selection.

7.4 Baseline L/S — ML Signal Screening

In-Sample:

Signal	Sharpe	Ann. Return	Ann. Vol	Max DD	Turnover Ann.
1 LightGBM_frs3 ★	0.601	5.09%	8.94%	-13.71%	1571%
3 XGBoost_frs3	0.560	4.65%	8.81%	-16.47%	1830%
4 PCA_Ridge_frs3	0.442	3.89%	9.71%	-18.05%	1675%
2 Ensemble_RankAvg_frs1	0.330	2.17%	7.31%	-16.72%	1815%
5 MLP_frs2	0.247	1.61%	7.66%	-14.06%	1745%

Out-of-Sample:

Signal	Sharpe	Ann. Return	Ann. Vol	Max DD	Turnover Ann.
2 Ensemble_RankAvg_frs1 ★	0.862	10.24%	12.16%	-4.59%	4633%
4 PCA_Ridge_frs3	0.267	2.47%	11.57%	-7.98%	3951%
5 MLP_frs2	-0.087	-1.27%	9.59%	-10.40%	4758%
3 XGBoost_frs3	-0.731	-4.79%	6.44%	-10.33%	2644%
1 LightGBM_frs3	-1.571	-8.54%	5.58%	-11.76%	2487%

LightGBM_frs3 (IS winner) collapses OOS (Sharpe -1.571). Ensemble (Signal 2) is the only ML signal clearly positive OOS. Single-model FRS3-label approaches overfit; blending and FRS1 labels generalise better.

8. Key Findings

Finding 01 — IS Winners Do Not Generalize OOS

Alpha#66 (IS BL Sharpe 0.967) collapses to Sharpe -0.483 OOS. Signal 1 / LightGBM_frs3 (IS BL Sharpe 0.601) collapses to -1.571 OOS. IS rank-order correlation with OOS is near zero, making IS-only selection unreliable for this universe.

Finding 02 — Alpha#23 is the Standout OOS Baseline Signal

Alpha#23 achieves OOS Baseline Sharpe 1.819, Ann. Return 19.34%, and a shallow Max DD of only -6.47% . Consistent across the LP signal-opt framework (OOS Sharpe 1.829). Its low IS Sharpe (0.432) means it would be rejected by IS-only selection — reinforcing the need for proper OOS evaluation.

Finding 03 — Signal 2 (Ensemble Rank Average) is the Most Robust ML Signal

The rank-average ensemble (FRS1 label) is the only ML signal with positive OOS Sharpe in both the Baseline L/S (0.862) and LP signal-opt (1.730) frameworks. FRS3-labeled single models generalize poorly OOS. Ensemble averaging and simpler return labels are more robust than complex single-model FRS3 forecasts.

Finding 04 — Alpha#57 Achieves the Highest OOS Sharpe in LP Mode

Alpha#57 reaches OOS LP Sharpe 2.174, Ann. Return 30.06% — the best single-signal result in the entire LP grid. It also performs well in BL mode (OOS Sharpe 1.315), suggesting consistent information content across both long-only and market-neutral contexts.

Finding 05 — Short-Only (SP) is Structurally Loss-Making

All SP Sharpes are ≤ 0 , reflecting the secular bull trend in US sector ETFs during the sample period. SP tracking remains useful for identifying short candidates for the Baseline L/S strategy, but standalone short-only portfolios are not viable here.

Finding 06 — Signal Quality is Framework-Specific

IS-strong signals in LP mode (Alpha#24, IS Sharpe 1.122) perform poorly in BL mode (IS Sharpe 0.119) and collapse OOS in BL (−0.375). A good long-only softmax signal is not automatically a good market-neutral signal. Both frameworks must be evaluated independently.

9. Next Steps

Priority	Task	Detail
● High	Signal Opt. Step 2 — Multi-Signal Bayesian Optimization	Linear mix $\sum \alpha_k g^{(k)}$ over Step 1 candidates; Bayesian opt + walk-forward CV; must beat Steps 0–1
● High	Baseline Parameter Tuning	Optimize stickiness threshold; add liquidity filter; evaluate borrow cost estimates
● Medium	Transaction Cost Sensitivity	Sweep <code>long_cost</code> over {0, 5, 10, 20} bps on Alpha#23, Alpha#57, Signal 2
● Medium	Walk-Forward OOS Validation	Rolling 12-month windows on best picks to confirm results are not period-specific
● Low	Signal Opt. Step 3 — Differentiable Neural Backtest	Neural network $\theta \rightarrow$ per-ETF logits $\rightarrow$ softmax $\rightarrow$ NAV; gradient through soft floor $\epsilon = 1e-6$

Limitation: All results are frictionless (zero fees, zero slippage). This is intentional — the goal is to evaluate signal predictive power in isolation. Transaction costs, particularly for high-turnover strategies, will materially reduce realised Sharpe. Cost sensitivity analysis is a planned next step.